e-Commerce consumer behaviour

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```
In [ ]:
```

1. Importing required libraries

```
In [1]: import numpy as np
        import pandas as pd

        import seaborn as sns
```

2. Loading the DataSet

```
In [2]: df = pd.read_csv('ECommerce_consumer behaviour.csv')
```

```
In [3]: df.head()
```

Out[3]:

	order_id	user_id	order_number	order_dow	order_hour_of_day	days_since_prior_order	product_id	add_to_cart_order	reordered	department_id	department	product_name
0	2425083	49125	1	2	18	NaN	17	1	0	13	pantry	baking ingredients
1	2425083	49125	1	2	18	NaN	91	2	0	16	dairy eggs	soy lactosefree
2	2425083	49125	1	2	18	NaN	36	3	0	16	dairy eggs	butter
3	2425083	49125	1	2	18	NaN	83	4	0	4	produce	fresh vegetables
4	2425083	49125	1	2	18	NaN	83	5	0	4	produce	fresh vegetables

3. Number of rows and columns

```
In [4]: df.shape
Out[4]: (2019501, 12)
```

4. DataFrame information

```
In [5]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2019501 entries, 0 to 2019500
Data columns (total 12 columns):
 #   Column                  Dtype
---  ------                  -----
 0   order_id                int64
 1   user_id                 int64
 2   order_number            int64
 3   order_dow               int64
 4   order_hour_of_day       int64
 5   days_since_prior_order  float64
 6   product_id              int64
 7   add_to_cart_order       int64
 8   reordered               int64
 9   department_id           int64
 10  department              object
 11  product_name            object
dtypes: float64(1), int64(9), object(2)
memory usage: 184.9+ MB
```

5. Checking missing values

```
In [6]: df.isna().sum()
Out[6]: order_id                       0
        user_id                        0
        order_number                   0
        order_dow                      0
        order_hour_of_day              0
        days_since_prior_order    124342
        product_id                     0
        add_to_cart_order              0
        reordered                      0
        department_id                  0
        department                     0
        product_name                   0
        dtype: int64
```

6. Checking duplicate values

```
In [7]: df.duplicated().sum()
Out[7]: 0
```

7. Number of unique values

```
In [8]: df.nunique()
Out[8]: order_id                  200000
        user_id                   105273
        order_number                 100
        order_dow                      7
        order_hour_of_day             24
        days_since_prior_order        31
        product_id                   134
        add_to_cart_order            137
        reordered                      2
        department_id                 21
        department                    21
        product_name                 134
        dtype: int64
```

8. Handle missing values in days_since_prior_order column

```
In [9]: df['days_since_prior_order']
Out[9]: 0          NaN
        1          NaN
        2          NaN
        3          NaN
        4          NaN
                  ...
        2019496    5.0
        2019497    3.0
        2019498    3.0
        2019499    3.0
        2019500    3.0
        Name: days_since_prior_order, Length: 2019501, dtype: float64
```

```
In [10]: df['days_since_prior_order'] = df['days_since_prior_order'].fillna('0')
```

```
In [11]: df['days_since_prior_order']
Out[11]: 0            0
         1            0
         2            0
         3            0
         4            0
                   ...
         2019496    5.0
         2019497    3.0
         2019498    3.0
         2019499    3.0
         2019500    3.0
         Name: days_since_prior_order, Length: 2019501, dtype: object
```

9. Converting days_since_prior_order into int data type

```
In [12]: df['days_since_prior_order'] = df['days_since_prior_order'].astype(int)
```

```
In [13]: df['days_since_prior_order']
Out[13]: 0          0
         1          0
         2          0
         3          0
         4          0
                   ..
         2019496    5
         2019497    3
         2019498    3
         2019499    3
         2019500    3
         Name: days_since_prior_order, Length: 2019501, dtype: int32
```

```
In [ ]:
```

```
In [14]: df.groupby('order_dow')['user_id'].agg(['count'])
```

Out[14]:

	count
order_dow	
0	391831
1	349236
2	261912
3	238730
4	234884
5	262157
6	280751

```
In [ ]:
```

10. groupby on order_dow(Day of the week) column

```
In [15]: df.groupby('order_dow')['user_id'].agg(['count'])
```

Out[15]:

	count
order_dow	
0	391831
1	349236
2	261912
3	238730
4	234884
5	262157
6	280751

```
In [16]: df.groupby('order_dow')['user_id'].agg(['count']).sort_values(by = 'count')
```

Out[16]:

	count
order_dow	
4	234884
3	238730
2	261912
5	262157
6	280751
1	349236
0	391831

```
In [17]: df.groupby('order_dow')['user_id'].agg(['count']).sort_values(by = 'count', ascending = False)
```

Out[17]:

	count
order_dow	
0	391831
1	349236
6	280751
5	262157
2	261912
3	238730
4	234884

```
In [18]: result = df.groupby('order_dow')['user_id'].agg(['count']).sort_values(by = 'count', ascending = False)
```

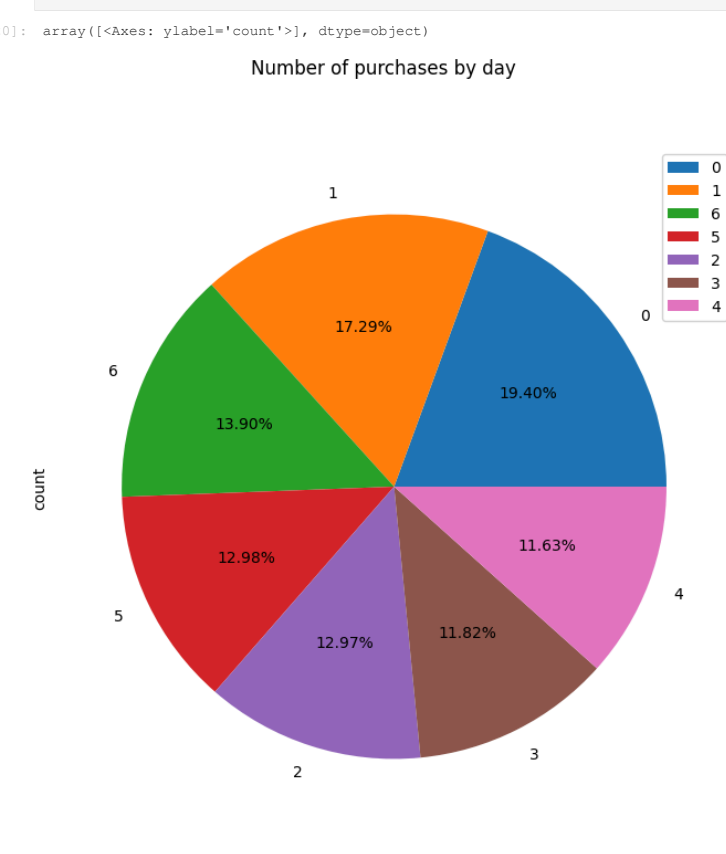
```
In [19]: result
```

Out[19]:

	count
order_dow	
0	391831
1	349236
6	280751
5	262157
2	261912
3	238730
4	234884

11. Number of purchases by day

```
In [20]: result.plot(
             kind = 'pie',
             autopct = '%1.2f%%',
             subplots = True,
             title = 'Number of purchases by day',
             figsize = (8, 8)
         )
Out[20]: array([<Axes: ylabel='count'>], dtype=object)
```



12. Time of day, when the order was made

```
In [21]: result1 = df.groupby('order_hour_of_day', as_index=False).agg({'user_id':'count'})
```

```
In [22]: result1
```

Out[22]:

	order_hour_of_day	user_id
0	0	13481
1	1	7283
2	2	4210
3	3	2994
4	4	3269
5	5	5732
6	6	18293
7	7	54143
8	8	106754
9	9	150248
10	10	173306
11	11	170291
12	12	163511
13	13	166376
14	14	167831
15	15	167157
16	16	158247
17	17	129383
18	18	102416
19	19	78516
20	20	62110
21	21	48857
22	22	40762
23	23	24331

```
In [23]: result1.sort_values(by='user_id',ascending=False)
```

Out[23]:

	order_hour_of_day	user_id
10	10	173306
11	11	170291
14	14	167831
15	15	167157
13	13	166376
12	12	163511
16	16	158247
9	9	150248
17	17	129383
8	8	106754
18	18	102416
19	19	78516
20	20	62110
7	7	54143
21	21	48857
22	22	40762
23	23	24331
6	6	18293
0	0	13481
1	1	7283
5	5	5732
2	2	4210
4	4	3269
3	3	2994

13. Keep the transactions separate like, morning, afternoon, evening, night

```
In [24]: def order_time(x):
             if x in [6,7,8,9,10,11,12]:
                 return 'Morning'
             if x in [13,14,15,16,17]:
                 return 'Afternoon'
             if x in [18,19,20,21,22]:
                 return 'Evening'
             if x in [23,24,0,1,2,3,4,5]:
                 return 'Night'
             return x
```

```
In [25]: df['order_time_list']=df['order_hour_of_day'].apply(order_time)
```

```
In [26]: df.head()
```

Out[26]:

	order_id	user_id	order_number	order_dow	order_hour_of_day	days_since_prior_order	product_id	add_to_cart_order	reordered	department_id	department	product_name	order_time_list
0	2425083	49125	1	2	18	0	17	1	0	13	pantry	baking ingredients	Evening
1	2425083	49125	1	2	18	0	91	2	0	16	dairy eggs	soy lactosefree	Evening
2	2425083	49125	1	2	18	0	36	3	0	16	dairy eggs	butter	Evening
3	2425083	49125	1	2	18	0	83	4	0	4	produce	fresh vegetables	Evening
4	2425083	49125	1	2	18	0	83	5	0	4	produce	fresh vegetables	Evening

```
In [27]: result2 = df.groupby('order_time_list')['user_id'].agg(['count'])
```

```
In [28]: result2
```

Out[28]:

	count
order_time_list	
Afternoon	788994
Evening	332661
Morning	836546
Night	61300

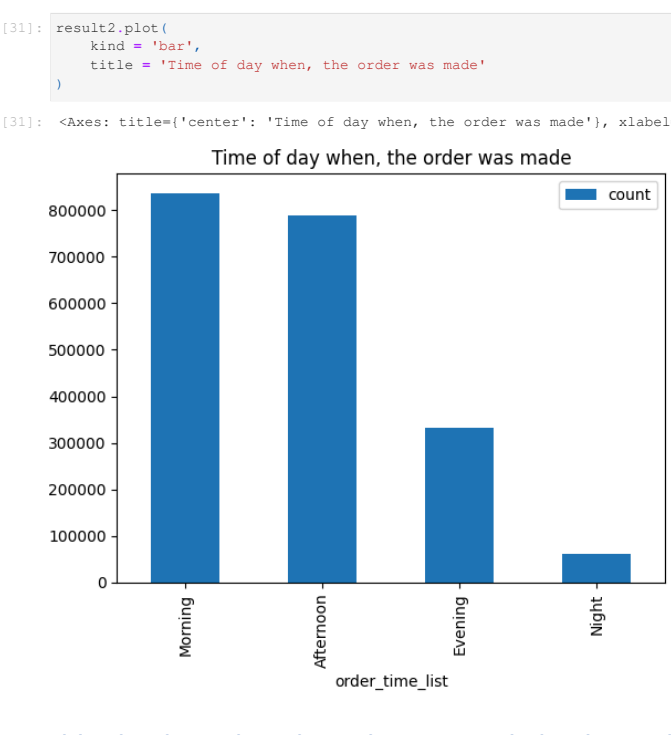
```
In [29]: result2 = result2.sort_values(by = 'count',ascending = False)
```

```
In [30]: result2
```

Out[30]:

	count
order_time_list	
Morning	836546
Afternoon	788994
Evening	332661
Night	61300

```
In [31]: result2.plot(
             kind = 'bar',
             title = 'Time of day when, the order was made'
         )
Out[31]: <Axes: title={'center': 'Time of day when, the order was made'}, xlabel='order_time_list'>
```



14. The time when the order was made by day and time of day

```
In [32]: df.pivot_table(
             index = 'order_dow',
             columns = 'order_time_list',
             values = 'user_id',
             aggfunc = ['count']
         )
```

Out[32]:

	count			
order_time_list	Afternoon	Evening	Morning	Night
order_dow				
0	162933	60195	157546	11157
1	130205	51825	158356	8850
2	100223	45497	108266	7926
3	94072	41737	95722	7199
4	89207	45359	92301	8017
5	101113	42311	110126	8607
6	111241	45737	114229	9544

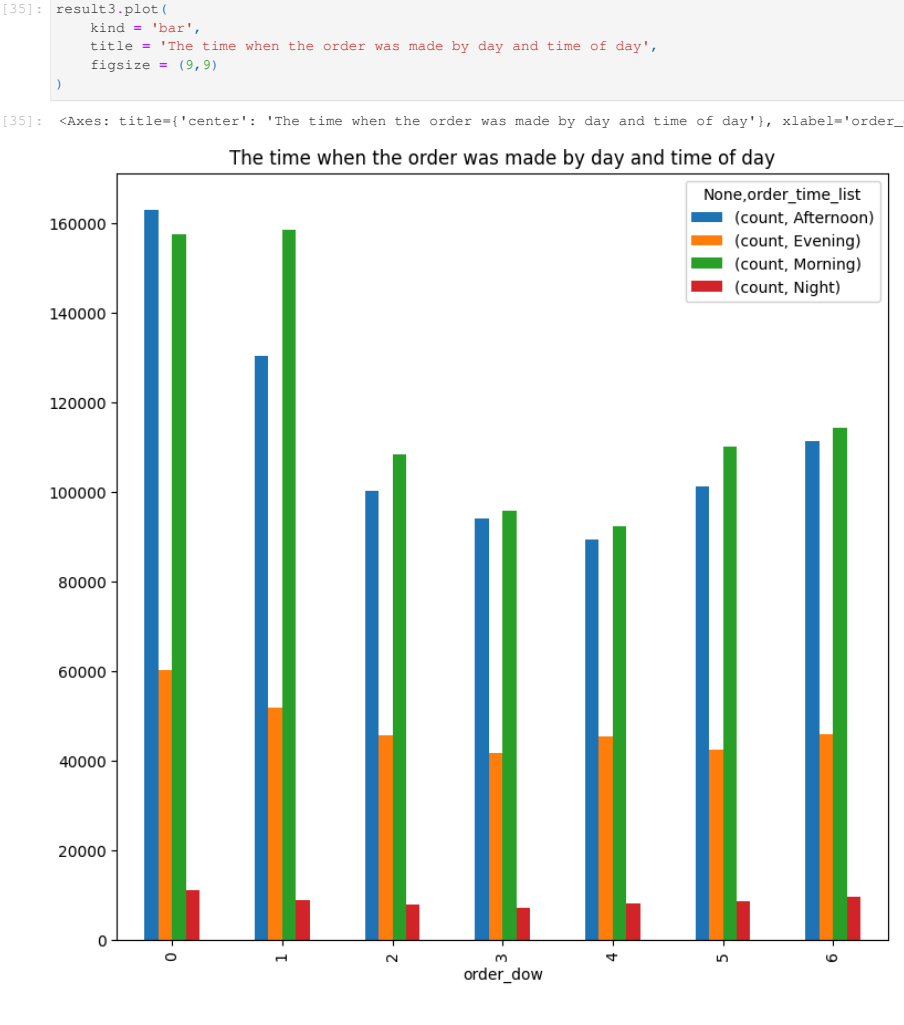
```
In [33]: result3 = df.pivot_table(
             index = 'order_dow',
             columns = 'order_time_list',
             values = 'user_id',
             aggfunc = ['count']
         )
```

```
In [34]: result3
```

Out[34]:

	count			
order_time_list	Afternoon	Evening	Morning	Night
order_dow				
0	162933	60195	157546	11157
1	130205	51825	158356	8850
2	100223	45497	108266	7926
3	94072	41737	95722	7199
4	89207	45359	92301	8017
5	101113	42311	110126	8607
6	111241	45737	114229	9544

```
In [35]: result3.plot(
             kind = 'bar',
             title = 'The time when the order was made by day and time of day',
             figsize = (9,9)
         )
Out[35]: <Axes: title={'center': 'The time when the order was made by day and time of day'}, xlabel='order_dow'>
```



15. The number of orders that users have made

```
In [36]: df.groupby('order_number')['user_id'].agg(['count'])
```

Out[36]:

	count
order_number	
1	124342
2	122397
3	120627
4	118573
5	105482
...	...
96	1051
97	834
98	964
99	716
100	493

100 rows × 1 columns

```
In [37]: def order_number_group(x):
             if x <= 20 :
                 x= '1-20 orders'
             elif x >20 and x <=40:
                 x= '21-40 orders'
             elif x >40 and x <=60:
                 x= '41-60 orders'
             elif x >60 and x <=80:
                 x= '61-80 orders'
             else:
                 x= '81-100 orders'
             return x
```

```
In [38]: df['order_number_group']=df['order_number'].apply(order_number_group)
```

```
In [39]: df
```

Out[39]:

	order_id	user_id	order_number	order_dow	order_hour_of_day	days_since_prior_order	product_id	add_to_cart_order	reordered	department_id	department	product_name	order_time_list	order_number_group
0	2425083	49125	1	2	18	0	17	1	0	13	pantry	baking ingredients	Evening	1-20 orders
1	2425083	49125	1	2	18	0	91	2	0	16	dairy eggs	soy lactosefree	Evening	1-20 orders
2	2425083	49125	1	2	18	0	36	3	0	16	dairy eggs	butter	Evening	1-20 orders
3	2425083	49125	1	2	18	0	83	4	0	4	produce	fresh vegetables	Evening	1-20 orders
4	2425083	49125	1	2	18	0	83	5	0	4	produce	fresh vegetables	Evening	1-20 orders
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
2019496	3390742	199430	16	3	18	5	83	8	0	4	produce	fresh vegetables	Evening	1-20 orders
2019497	458285	128787	42	2	19	3	115	1	1	7	beverages	water seltzer sparkling water	Evening	41-60 orders
2019498	458285	128787	42	2	19	3	32	2	1	4	produce	packaged produce	Evening	41-60 orders
2019499	458285	128787	42	2	19	3	32	3	1	4	produce	packaged produce	Evening	41-60 orders
2019500	458285	128787	42	2	19	3	123	4	1	4	produce	packaged vegetables fruits	Evening	41-60 orders

2019501 rows × 14 columns

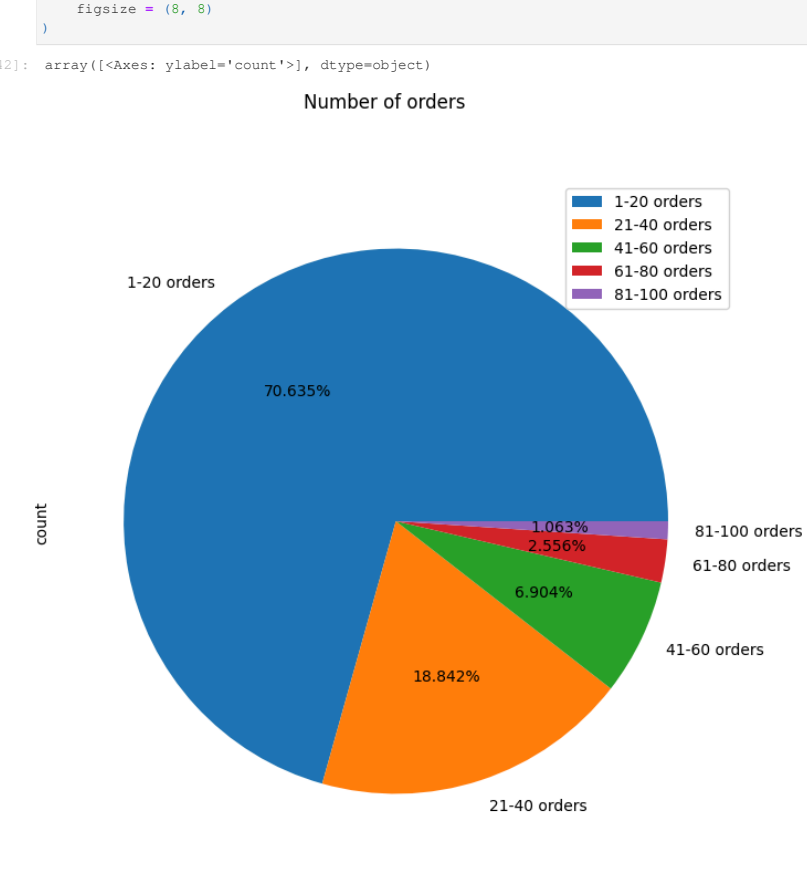
```
In [40]: result4 = df.groupby('order_number_group')['user_id'].agg(['count']).sort_values(by = 'count', ascending = False)
```

```
In [41]: result4
```

Out[41]:

	count
order_number_group	
1-20 orders	1426484
21-40 orders	380514
41-60 orders	139421
61-80 orders	51612
81-100 orders	21470

```
In [42]: result4.plot(
             kind = 'pie',
             autopct = '%1.3f%%',
             subplots = True,
             title = 'Number of orders',
             figsize = (8, 8)
         )
Out[42]: array([<Axes: ylabel='count'>], dtype=object)
```

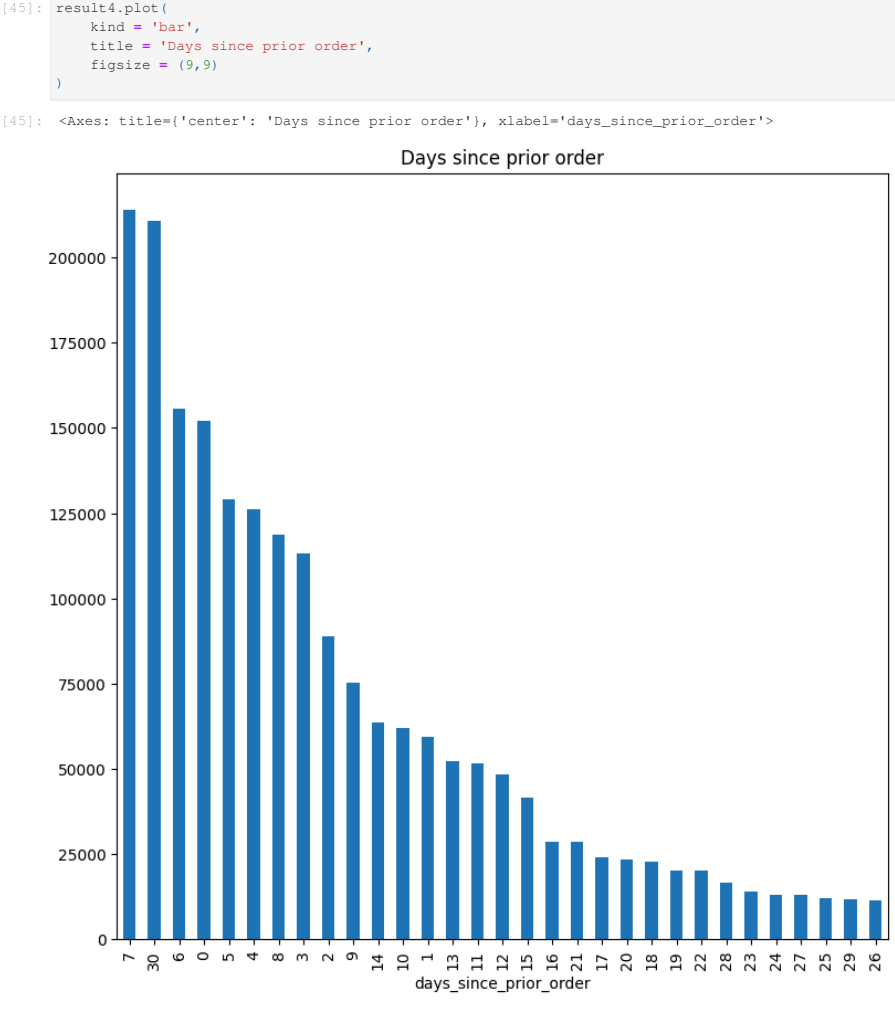


16. How many days have passed since the pre-order

```
In [43]: result4 = df.groupby('days_since_prior_order')['user_id'].agg('count').sort_values(ascending = False)
```

```
In [44]: result4
Out[44]: days_since_prior_order
         7     214126
         30    210814
         6     155685
         0     152015
         5     129089
         4     126250
         8     118722
         3     113263
         2      88737
         9      75120
         14     63414
         10     62073
         1      59200
         13     52204
         11     51476
         12     48171
         15     41531
         16     28632
         21     28359
         17     23921
         20     23234
         18     22777
         19     20140
         22     19950
         28     16459
         23     13779
         24     12833
         27     12804
         25     11832
         29     11590
         26     11301
         Name: user_id, dtype: int64
```

```
In [45]: result4.plot(
             kind = 'bar',
             title = 'Days since prior order',
             figsize = (9,9)
         )
Out[45]: <Axes: title={'center': 'Days since prior order'}, xlabel='days_since_prior_order'>
```



17. Top 15 most popular products from users

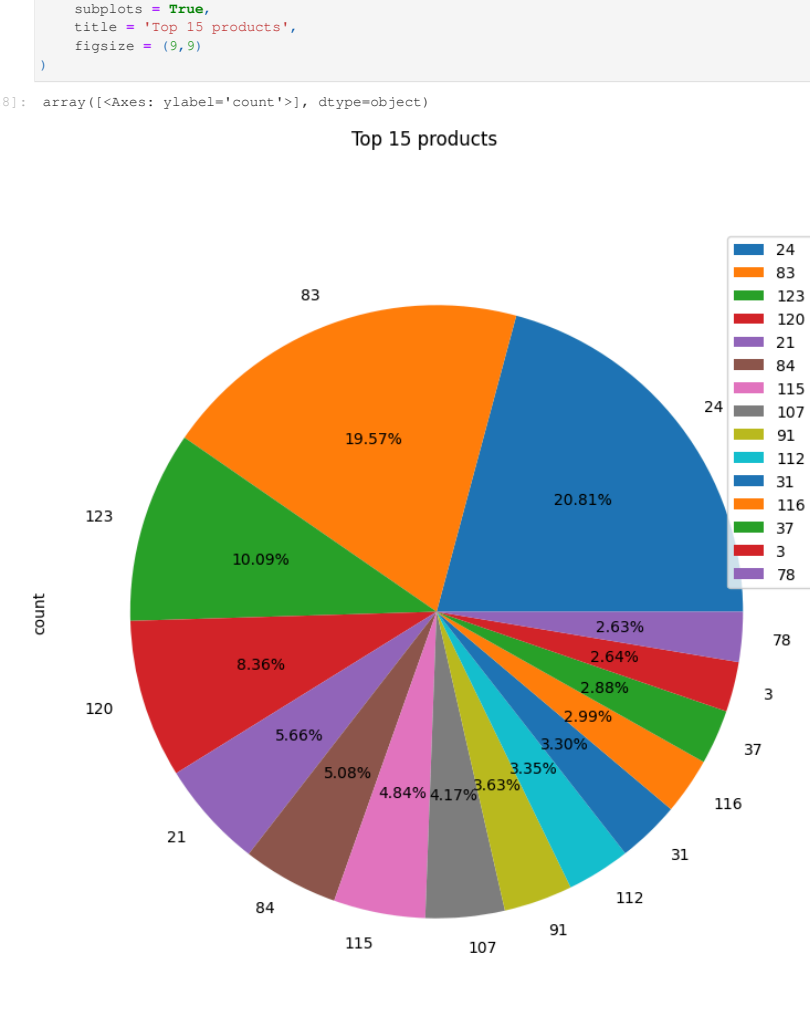
```
In [46]: result5 = df.groupby('product_id')['user_id'].agg(['count']).sort_values(by = 'count',ascending = False).head(15)
```

```
In [47]: result5
```

Out[47]:

	count
product_id	
24	226039
83	212611
123	109596
120	90751
21	61502
84	55150
115	52564
107	45306
91	39389
112	36381
31	35893
116	32432
37	31323
3	28639
78	28574

```
In [48]: result5.plot(
             kind = 'pie',
             autopct = '%1.2f%%',
             subplots = True,
             title = 'Top 15 products',
             figsize = (9,9)
         )
Out[48]: array([<Axes: ylabel='count'>], dtype=object)
```



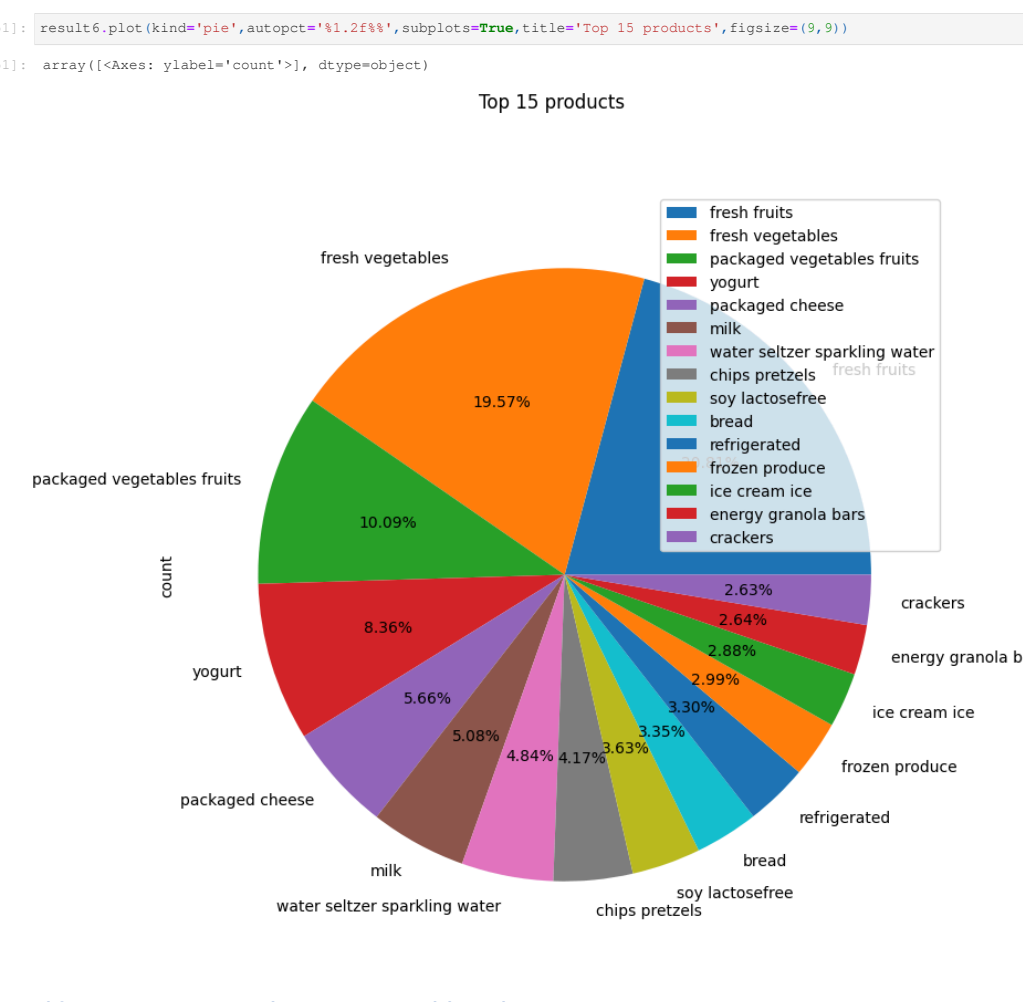
```
In [49]: result6 = df.groupby('product_name')['user_id'].agg(['count']).sort_values(by = 'count',ascending = False).head(15)
```

```
In [50]: result6
```

Out[50]:

	count
product_name	
fresh fruits	226039
fresh vegetables	212611
packaged vegetables fruits	109596
yogurt	90751
packaged cheese	61502
milk	55150
water seltzer sparkling water	52564
chips pretzels	45306
soy lactosefree	39389
bread	36381
refrigerated	35893
frozen produce	32432
ice cream ice	31323
energy granola bars	28639
crackers	28574

```
In [51]: result6.plot(kind='pie',autopct='%1.2f%%',subplots=True,title='Top 15 products',figsize=(9,9))
Out[51]: array([<Axes: ylabel='count'>], dtype=object)
```



18. How many products users add to the cart

```
In [52]: atco_u = df.groupby('add_to_cart_order', as_index = False).agg({'user_id': 'count'}).sort_values(by='user_id',ascending=False)
```

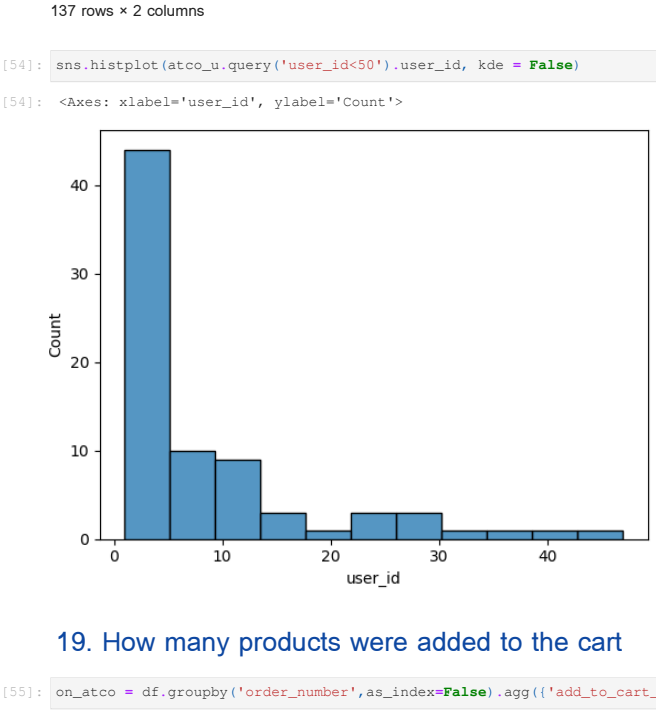
```
In [53]: atco_u
```

Out[53]:

	add_to_cart_order	user_id
0	1	200000
1	2	190134
2	3	178480
3	4	165743
4	5	151983
...	...	...
112	113	1
111	112	1
110	111	1
109	110	1
136	137	1

137 rows × 2 columns

```
In [54]: sns.histplot(atco_u.query('user_id<50').user_id, kde = False)
Out[54]: <Axes: xlabel='user_id', ylabel='Count'>
```



19. How many products were added to the cart

```
In [55]: on_atco = df.groupby('order_number',as_index=False).agg({'add_to_cart_order': 'count'}).sort_values(by='add_to_cart_order', ascending = False)
```

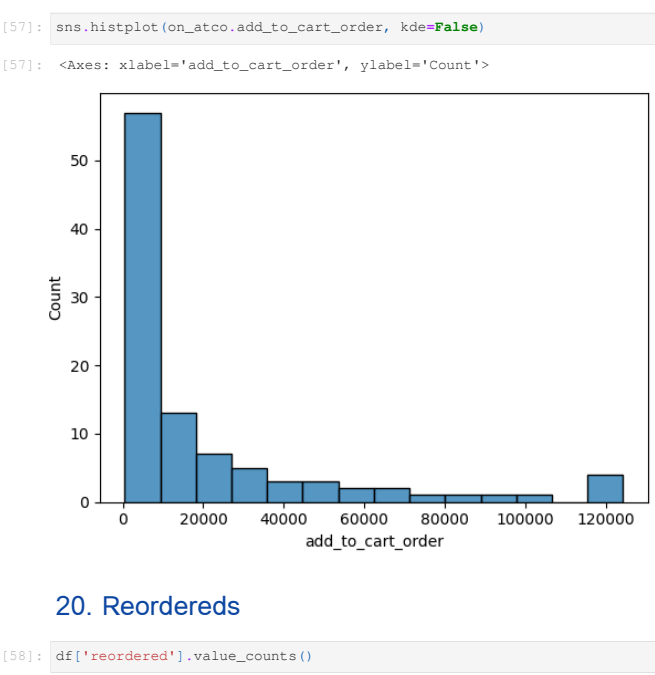
```
In [56]: on_atco
```

Out[56]:

	order_number	add_to_cart_order
0	1	124342
1	2	122397
2	3	120627
3	4	118573
4	5	105482
...	...	...
96	97	834
90	91	826
94	95	741
98	99	716
99	100	493

100 rows × 2 columns

```
In [57]: sns.histplot(on_atco.add_to_cart_order, kde=False)
Out[57]: <Axes: xlabel='add_to_cart_order', ylabel='Count'>
```



20. Reordereds

```
In [58]: df['reordered'].value_counts()
Out[58]: reordered
         1    1190986
         0     828515
         Name: count, dtype: int64
```

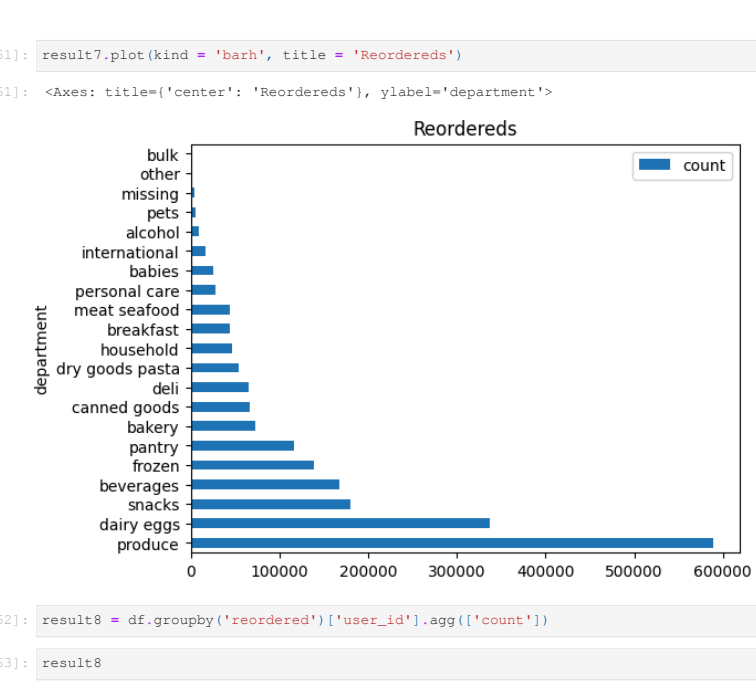
```
In [59]: result7 = df.groupby('department')['reordered'].agg(['count']).sort_values(by='count',ascending=False)
```

```
In [60]: result7
```

Out[60]:

	count
department	
produce	588996
dairy eggs	336915
snacks	180692
beverages	168126
frozen	139536
pantry	116262
bakery	72983
canned goods	66053
deli	65176
dry goods pasta	54054
household	46446
breakfast	44605
meat seafood	44271
personal care	28134
babies	25940
international	16738
alcohol	9439
pets	6013
missing	4749
other	2240
bulk	2133

```
In [61]: result7.plot(kind = 'barh', title = 'Reordereds')
Out[61]: <Axes: title={'center': 'Reordereds'}, ylabel='department'>
```



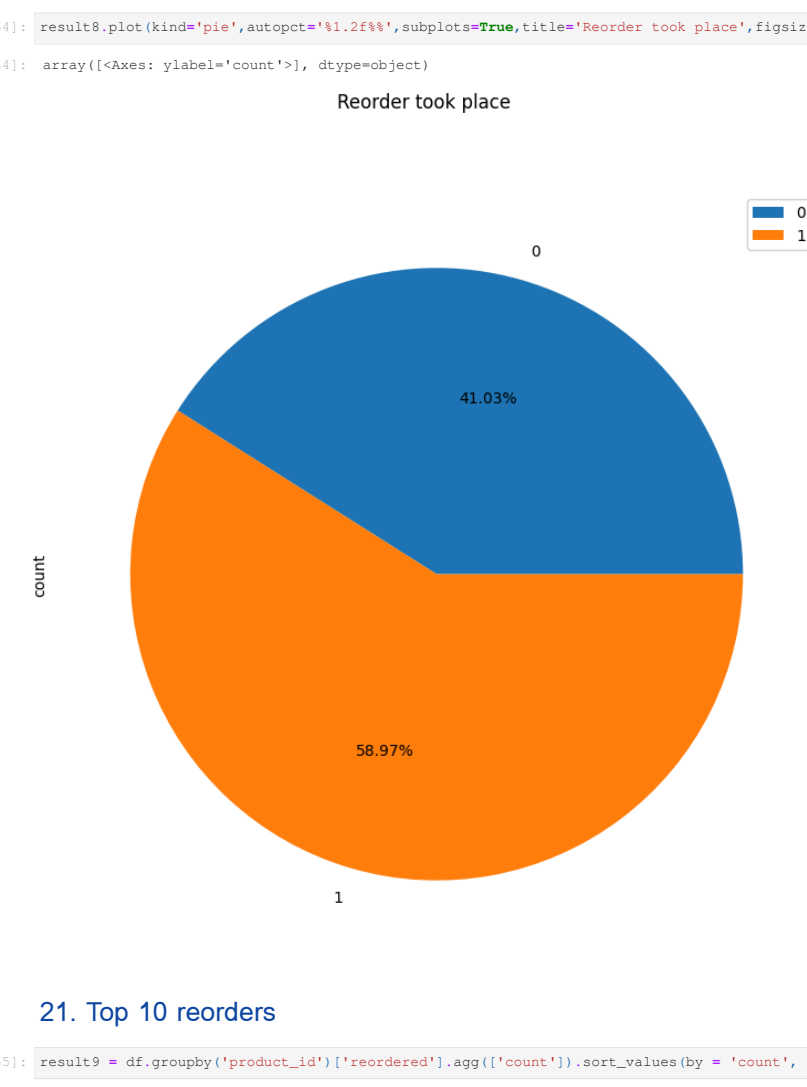
```
In [62]: result8 = df.groupby('reordered')['user_id'].agg(['count'])
```

```
In [63]: result8
```

Out[63]:

	count
reordered	
0	828515
1	1190986

```
In [64]: result8.plot(kind='pie',autopct='%1.2f%%',subplots=True,title='Reorder took place',figsize=(9,9))
Out[64]: array([<Axes: ylabel='count'>], dtype=object)
```



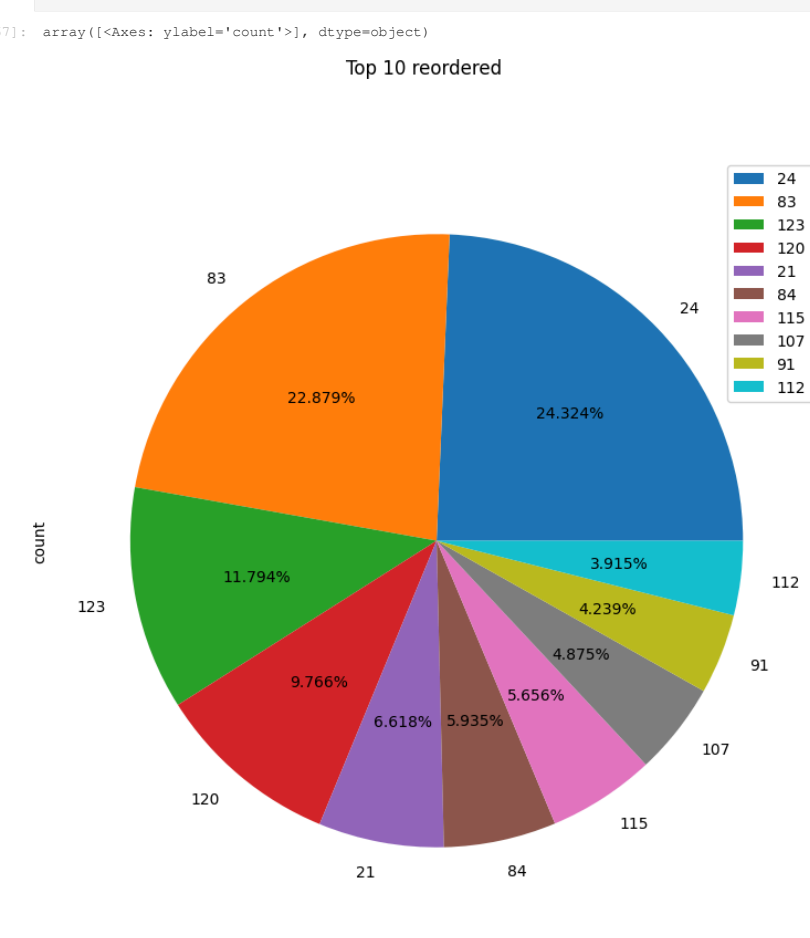
21. Top 10 reorders

```
In [65]: result9 = df.groupby('product_id')['reordered'].agg(['count']).sort_values(by = 'count', ascending = False).head(10)
```

```
In [66]: result9
```

Out[66]:

	count
product_id	
24	226039
83	212611
123	109596
120	90751
21	61502
84	55150
115	52564
107	45306
91	39389
112	36381

```
In [67]: result9.plot(
             kind = 'pie',
             autopct = '%1.3f%%',
             subplots = True,
             title = 'Top 10 reordered',
             figsize = (9, 9)
         )
Out[67]: array([<Axes: ylabel='count'>], dtype=object)
```